MIL-M-60003D(MI)
8 March 1983
SUPERSEDING
MIL-M-60003C(MI)
19 October 1973

# MILITARY SPECIFICATION

## MOTOR, ROCKET, ASSEMBLY, M54 OR M123

## PARTS FOR

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally required by the drawings, and provides quality assurance provisions for the fabrication, assembly and packing of one type of rocket motor parts assembly designated as Motor, Rocket, Assembly, M54 or M123.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | Ammunition and Special Weapons; General Specification for |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

FSC-1340

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| | |
|---|---|
| MIL-A-8625 | Anodic Coatings for Aluminum and Aluminum Alloy |

STANDARDS

MILITARY

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|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-1168 | Lot Numbering of Ammunition |
| MIL-STD-1169 | Packaging, Packing and Marking for Shipment of Inert Ammunition Components |
| MIL-STD-1235 | Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

| | |
|---|---|
| 10048509 | Rocket Motor, M54 |
| 11508955 | Rocket Motor Assy, M123 |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| ASTM-B117 | Salt Spray (Fog) Testing, Standard Method of |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).



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Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

2.3 Order of precedence. In the event a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials used in the fabrication of the motor shall be in accordance with applicable drawings and specifications.

3.2 Assemblies. The motor assemblies shall comply with all requirements specified on Drawing (dwg.) 10048509, or 11508955 and with all requirements specified on applicable specifications.

3.3 Embrittlement. Fin springs manufactured for use in the assembly shall be free from embrittlement as evidenced by not breaking when extended or distorted to the point of minimum permanent distortion.

3.4 Heat treatment of motor bodies.

3.4.1 Maximum Heat-Treatment Temperatures. The material used in the fabrication of the motor bodies and the finished motor bodies shall at no time be subjected to Fahrenheit temperatures in excess of 880 degrees for 7001 alloy and 890 degrees for 7050 alloy. If a salt bath furnace is utilized for solution heat-treatment, the cold-worked motor bodies shall be pre-heated to relieve internal stresses to preclude the development of cracks during the normal rapid rise in temperature upon introduction into the salt bath.

3.4.2 Grain structure. The motor bodies, after all heat-treatments and aging treatments have been completed, shall not contain eutectic melting, harmful grain boundary conditions or Banding (segregation of inter-metallic compounds) when examined under 200 to 500 power magnification.

3.5 Physical strength of motor body.

3.5.1 Yield strength. The portion of the motor body from the threaded end to the nozzle throat, shall not rupture; the threads shall not fail to hold the motor body in the test fixture, and the motor body shall conform to the drawing dimensions after having been subjected to an internal hydrostatic pressure of 14,000 plus 200 or minus 500 pounds per square inch. The pressure shall be applied while the motor body is not restrained or confined so as to restrict deformation, except at the threads, where deformation will be limited by the threads of the test fixture and the threads in the test fixture shall represent the minimum metal conditions of the closure.



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3.5.2 Ultimate strength. The motor body, from the threaded end to the nozzle throat, shall not rupture nor shall the threads fail while an internal hydrostatic pressure of 14,800 pounds per square inch minimum is applied in such a manner as not to restrict deformation.

3.6 Protective coating. The exterior and interior surfaces of the motor body, after application of the protective coating, shall not contain in excess of seven (7) total isolated bare spots or pits larger than one thirty-second (1/32) of an inch in diameter after the finished motor body has been exposed to a salt spray solution containing five (5) percent salt for a minimum of one hundred sixty eight (168) hours. Fins shall not contain in excess of three (3) total bare spots or pits in excess of one thirty second (1/32) of an inch in diameter after exposure to an identical environment for the same period of time as the motor body.

3.7 Assemblies.

3.7.1 Fin action. Fins shall move freely, shall not bend or twist, and shall spring open without hesitation. All fins shall open to within the limits specified on the applicable drawing. Springs, lugs and eyelets shall show no evidence of distortion, or disassembly, and eyelets shall not be cracked or split in excess of the limits specified on the applicable drawing.

3.7.2 Physical strength of exit cone. The motor body shall not crack or deform beyond drawing tolerances when stressed within the following limits at the locations specified. Stress readings shall indicate the hoop stress to which the motor exit cone has been subjected. Maximum stress shall not exceed 80,000 pounds per square inch. Minimum stress at all unspecified exit cone locations shall be that resulting from the application of the specified test equipment, to achieve the minimum stress levels specified at the specified locations. Location of stress measurement points shall be measured along the longitudinal axis of the motor body beginning at the lip of the exit cone. Stress measurement points and minimum stress levels shall be:

| | |
|---|---|
| 3/16 inch | 29,000 pounds per square inch |
| 1 inch | 16,000 pounds per square inch |
| 1 3/8 inch | 34,000 pounds per square inch |

At and beyond the throat and through the remainder of the motor body, stress shall be that resulting from 10,000 pounds per square inch minimum to 13,000 pounds per square inch maximum internal pressure.

3.7.3 Defects, physical. The nozzle exit cone area of the motor body shall be free from all external and internal cracks and metallic discontinuities when examined by "eddy current" equipment adjusted to a sensitivity to disclose metallic discontinuities of 0.062 inch maximum diameter hole.



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3.7.4 Defects, visual. The motor bodies, after all assembly and machining operations and all other tests, shall be free from deformation beyond drawing tolerances, cracks, surface seams, laps, scratches in excess of .002 inch deep, or any other metallic discontinuities when visually examined under three (3) power magnification.

3.7.5 Static firing. The motor bodies shall not fracture or fragment when statically tested with production propellant charges and igniters at 150 degrees Fahrenheit, minimum.

3.8 Workmanship. All parts and assemblies shall be free of chips, dirt, grease, rust, scale and other foreign material. The cleaning method shall not be injurious to any or the parts, nor shall the parts be contaminated by the cleaning agents. All markings shall be clean and sharply defined.

3.9 Submission of first article sample. This specification makes provisions for the submission of a first article sample for evaluation by the Government. Procedures for submission of the first article samples shall be as specified in the contract (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.



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c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission. Prior to initiating regular production, the contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (see 6.4) The first article sample shall consist of:

200 complete assemblies for static firing at 150°F prior to physical strength test of exit cone.

25 motor bodies that have not been subjected to hydrotesting or anodizing.

10 sets of all components. Motor bodies in this set shall be after hydrotest and anodizing.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3.1 of this specification and any or all of the requirements of the applicable drawing.



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4.2.3 Rejection. The first article sample shall be rejected if any component or assembly fails to comply with any of the applicable requirements referenced in 4.3.2 and 4.3.3.1. The first article sample shall also be rejected if one or more specimens are classed defective during the ultimate-strength test of motor bodies (see 4.3.3.1.4) or if the average burst or failure level of the combined sample is less than 16,500 pounds per square inch. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Quality conformance (production lot acceptance) inspection.

4.3.1 Lot formation. The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. The examination listed in the following subparagraphs and 4.4.3.2 shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects shall be performed 100 percent. Certain Major Defects, when so specified, shall be examined 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph, except where 100 percent inspection is specified. This option shall be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.



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4.3.2.1 Fin (see dwg. 10048520 or 11508951 covering a detail of dwg. 10048509 or 11508955).

| Categories | Defects | Method of Inspection | Code No. (See 6.2) |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 Percent | | |
| 101. | Thickness, minimum (min.)..... | Gage | 01001 |
| 102. | Diameter of hole, maximum (max.)..... | Gage | 01002 |
| Minor: | AQL 0.65 Percent | | |
| 201. | Evidence of poor workmanship (see 3.8)..... | Visual | 01003 |

4.3.2.2 Body, motor rocket: M54 or M123, after anodizing (see dwg. 10048519 (M54) or 11508954 (M123) covering a detail of dwg. 10048509 (M54) or 11508955 (M123).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Hydrostatic test stamp missing (see 3.5.1 and 6.5)..... | Visual | 02001* |
| Major: | 100% Inspection | | |
| 101. | Concentricity of thread with inside diameter..... | Gage | 02002 |
| 102. | Concentricity of throat with inside diameter..... | Gage | 02003 |
| 103. | Concentricity of inside of exit cone with inside diameter..... | Gage | 02004 |
| 104. | Longitudinal alignmentof slots with centerline of inside diameter..... | Gage | 02005 |
| 106. | Pitch diameter, min. ..... | Gage | 02031 |
| 107. | Length of thread, min. ..... | Gage | 02008 |
| 108. | Lengths of datum diameters in exit cone section..... | Gage | 02009 |
| 109. | Wall thickness of cylindrical section, min. ..... | Gage | 02010 |
| 110. | Wall thickness at end of exit cone, min. ..... | Gage | 02011 |
| 111. | Wall thickness at throat, min..... | Gage | 02012 |



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| 112. | Inside diameter of cylindrical section, min. ..... | Gage | 02013 |
| 113. | Inside diameter at throat..... | Gage | 02014 |
| 114. | Diameter of eyelet holes, max. ..... | Gage | 02015 |
| 115. | Distance from centerline of holes to bottom of slots, min. ..... | Gage | 02016 |
| 116. | Width of slots, max. ..... | Gage | 02017 |
| 117. | Total length..... | Gage | 02018 |
| 118. | Protective coating missing, incomplete or damaged (except for hanger marks)..... | Visual | 02019 |
| 119. | Major diameter, min..... | Gage | 02032 |
| Minor: | AQL 1.00 Percent | | |
| 201. | Chamber missing at inside diameter of threaded end..... | Visual | 02020 |
| 202. | Evidence of poor workmanship (see 3.8)..... | Visual | 02021 |

4.3.2.3 Spring, fin (see dwg. 10048518, covering a detail of dwg. 10048509 (M54) or 11508955 (M123).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.40 Percent | | |
| 101. | Torque inadequate at either angle of deflection..... | Gage | 03001 |
| Minor: | AQL 0.65 Percent | | |
| 201. | Evidence of poor workmanship (see 3.8)..... | Visual | 03002 |



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4.3.2.4 Assembly (see dwg. 10048509 (M54) or 11508955 (M123).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | | | |
| 1. | Exit cone physical strength test stamp missing (see 3.7.2)..... | Visual | 04001* |
| 2. | Exit cone physical defects test stamp missing (see 3.7.3)..... | Visual | 04002* |
| Major: | AQL 0.65 Percent | | |
| 101. | Fin action (see 3.7.1 and 4.4.9)..... | Test | 04003 |
| 102. | Weight, max. ..... | Gage | 04004 |
| 103. | Diameter across folded fins, max. ..... | Gage | 04005 |
| 104. | Components insecure, damaged (see 3.7.4), missing or incorrectly assembled..... | Visual | 04006 |
| Minor: | AQL 1.00 Percent | | |
| 201. | Marking missing, incorrect or illegible..... | Visual | 04007 |
| 202. | Evidence of poor workmanship (see 3.8)..... | Visual | 04008 |

4.3.3 Testing.

4.3.3.1 First article sample. The first article sample shall be examined 100 percent for all of the characteristics in 4.3.2, (without regard to classification and AQL) and then subjected to the tests listed below. Inspection and test equipment shall conform to the requirements of 4.3.4. Rejection of the first article sample shall be in accordance with 4.2.3.

4.3.3.1.1 Embrittlement (see 3.3) - Code No. 03003. Ten (10) of the fin springs shall be tested as specified in 4.4.1.

4.3.3.1.2 Grain structure in motor body (see 3.4.1 and 3.4.2) - Code No. 02022. Five (5) of the motor bodies from the first article sample that were not hydrotested or anodized shall be tested as specified in 4.4.2.

4.3.3.1.3 Yield strength of motor body (see 3.5.1) - Code No. 02023. The remaining twenty (20) motor bodies from the first article sample that were not hydrotested or anodized shall be tested as specified in 4.4.3.

4.3.3.1.4 Ultimate strength of motor body (see 3.5.2) Code No. 02024. The twenty (20) motor bodies from 4.3.3.1.3 shall be tested as specified in 4.4.4. Samples subjected to this test shall be scrapped.



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4.3.3.15 Throat contour of motor body (see dwg. 10048519 (M54) or 11508954 (M123) Code No. 02025. The ten (10) motor bodies from the sets of components (hydrotested and anodized) shall be tested as specified in 4.4.5.

4.3.3.1.6 Surface finish of motor body (see dwg. 10048519 (M54) or 11508954 (M123) Code No. 02026. The ten (10) motor bodies from 4.3.3.1.5 shall be examined for surface finish as specified in 4.4.6.

4.3.3.1.7 Protective coating of motor body and fins.

4.3.3.1.7.1 Salt spray method (see 3.6); fin, Code No. 01004; and Motor body, code No. 02027. Six (6) motor bodies from 4.3.3.1.6 and six (6) fins from the first article sample shall be tested as specified in 4.4.7.

4.3.3.1.7.2 Direct measurement method (see dwg. 10048519 (M54) or 11508954 (M123) and dwg. 10048520 (M54) or 11508951 (M123) fin, Code No. 01005; and motor body, Code No. 02028. Four (4) of the motor bodies from 4.3.3.1.6 and four (4) of the fins from the first article sample shall be tested as specified in 4.4.8.

4.3.3.1.8 Physical strength of exit cone (see 3.7.2) Code No. 04009. The two hundred (200) complete assemblies shall be tested as specified in 4.4.10.

4.3.3.1.9 Physical defects in motor body (see 3.7.3) Code No. 04010. The two hundred (200) complete assemblies from 4.3.3.1.8 shall be tested as specified in 4.4.11.

4.3.3.1.10 Visual defects in motor body (see 3.7.4) Code No. 04011. The two hundred (200) complete assemblies from 4.3.3.1.9 shall be examined as specified in 4.4.12.

4.3.3.1.11 Static firing of assembly (see 3.7.5) Code No. 04012. The two hundred (200) complete assemblies from 4.3.3.1.10 shall be tested as specified in 4.4.13.

4.3.3.2 Production sample.

4.3.3.2.1 Embrittlement (see 3.3) major defect, Code No. 03003. A sample of ten (10) fin springs shall be selected from each heat-treatment batch and tested as specified in 4.4.1. The heat-treatment batch shall be rejected if one or more springs are classed defective.

4.3.3.2.2 Grain structure and maximum heat-treatment temperature in motor body (see 3.4.1 and 3.4.2), critical defect, Code No. 02022* and excessive heat-treatment temperature, Code No. 02030*. A sample of two (2) motor bodies shall be selected from each heat-treatment and age-treatment batch and tested as specified in 4.4.2. The batch shall be rejected if one or more bodies are classed defective, or if the temperature recording equipment indicates that the upper temperature limit has been exceeded during any phase of fabrication.



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4.3.3.2.3 Yield strength of motor body (see 3.5.1), critical defect, code No. 02023*. This test shall be performed one hundred (100) percent. The test shall be performed as specified in 4.4.3. Motor bodies classed defective shall be rejected and removed from the lot.

4.3.3.2.4 Ultimate strength of motor body (see 3.5.2), Code No. 02024; FOR INFORMATION ONLY. A sample of twenty (20) motor bodies shall be selected from each motor body inspection lot (after satisfactory conformance to 4.3.3.2.3) and tested as specified in 4.4.4. The test results shall be recorded and copies furnished as specified by the contract (see 6.1). Samples subjected to this test shall be scrapped.

4.3.3.2.5 Throat contour of motor body (see dwg. 10048519 (M54) or 11508954 (M123) major defect, Code No. 02025. A sample of thirty (30) motor bodies shall be selected from each inspection lot and tested as specified in 4.4.5. The inspection lot shall be rejected if two (2) or more bodies are classed defective. If one (1) body is classed defective, a second sample of thirty (30) shall be selected from the inspection lot and tested as specified in 4.4.5. Testing for Government acceptance may be stopped and the inspection lot shall be rejected upon encountering (1) defective motor body in this second sample.

4.3.3.2.6 Surface finish of motor body (see dwg. 10048519 (M54) or 11508954 (M123) major defect, Code No. 02026. A sample of thirty two (32) motor bodies shall be selected from each inspection lot and tested as specified in 4.4.6. The inspection lot shall be rejected if two (2) or more bodies are classed defective. If one (1) body is classed defective, a second sample of thirty-two (32) motor bodies shall be selected from the inspection lot and tested as specified in 4.4.6. Testing for Government acceptance may be stopped and the inspection lot shall be rejected upon encountering one (1) defective motor body in this second sample.

4.3.3.2.7 Protection against atmospheric stress corrosion (see dwg. 10048519 (M54) or 11508954 (123) critical defect, Code No. 02029*. The contractor shall establish process controls and schedules to facilitate the Government's surveillance of the contractor's conformance to assure against exceeding the allowable elapsed time from when any surface of the motor is final machined until the protective coating is applied. All products not adequately protected shall be rejected and removed from the process and scrapped.

4.3.3.2.8 Protective coating of fins and motor body. In determining the adequacy of the protective finish, the contractor may select either of the following methods for acceptance testing of regular production of fins and motor bodies for the Government.



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4.3.3.2.8.1 Salt spray method (see 3.6) major defects, fin, Code No. 01004; and motor body, Code No. 02027. A sample of three (3) motor bodies and three (3) fins shall be selected from each inspection lot and tested as specified in 4.4.7. The applicable inspection lot shall be rejected if one or more of the motor bodies or one or more fins are classed defective.

4.3.3.2.8.2 Direct measurement method (see dwg. 10048519 (M54) or 11508954 (M123) and dwg. 10048520 (M54) or 11508951 (M123) major defects; fin, Code No. 01005; and Motor Body, Code No. 02028. A sample of three (3) motor bodies and one (1) fin shall be selected from each inspection lot and tested as specified in 4.4.8. The inspection lot shall be rejected if one or more motor bodies are classed defective, or if the fin is classed defective.

4.3.3.2.9 Physical strength of exit cone (see 3.7.2) critical defect, Code No. 04009*. This test shall be performed one hundred (100 percent on final assemblies after completion of all fabrication procedures and all testing, except the testing listed below. This test shall be performed as specified in 4.4.10. Assemblies classed defective shall be rejected and removed from the lot.

4.3.3.2.10 Physical defects in motor body (see 3.7.3) critical defect, Code No. 04010*. This test shall be performed one hundred (100) percent on the product acceptable in 4.3.3.2.9. This test shall be performed as specified in 4.4.11. Assemblies classed defective shall be rejected and removed from the lot.

4.3.3.2.11 Visual defects in motor body (see 3.7.4) critical defect, Code No. 04012*. This examination shall be performed one hundred (100) percent on the product acceptable in 4.3.3.2.10. This examination shall be performed as specified in 4.4.12. Assemblies classed defective shall be rejected and removed from the lot.

4.3.3.2.12 Static firing of assembly (see 3.7.5) critical defect, Code No. 04013. A sample of two hundred (200) complete assemblies shall be selected from each lot and tested at a Government Proving Ground (see 6.1) as specified in 4.4.13. The lot shall be rejected and testing for Government acceptance shall be stopped upon encountering one assembly that is classed defective.

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests required by 4.3.2 and 4.3.3 of this specification shall be designed and fabricated by the contractor. Refer to 6.3 for details concerning responsibilities for inspection equipment design and Government approval.



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4.4 Test methods and procedures. The test methods and procedures described in each of the following tests represent only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using these or substitute methods and procedures. Whichever option is selected, detailed descriptions must be submitted to the Government for approval prior to use. Government approval will be based on an evaluation and demonstration to determine the effectiveness of all of the tests and test procedures.

4.4.1 Embrittlement. The fin spring shall be torqued in a manner simulating the actual use of the spring. The spring helix shall be placed over a steel rod 0.124 plus 0.000 minus 0.010 inch in diameter and the torque applied and removed to affect increasing angles of deflection until the spring fails to return to its original configuration (which is the definition of minimum permanent deformation). The spring shall be classed defective if it breaks or cracks prior to achieving the condition of minimum permanent deformation. Samples subjected to this test shall be scrapped and not returned to any lot.

4.4.2 Grain structure and maximum temperature. The contractor shall provide automatic time and temperature recording equipment to monitor the applicable sequences of heat-treatment to assure compliance with the specified requirements for maximum temperature. To determine presence of micorscopic defects, the motor body shall be sectioned through diametrically opposed fin slots and the two halves separated from the cylindrical portion of the body at a point to include approximately one half (1/2) inch of the cylindrical section with the throat and exit cone sections. The cylindrical sections shall be cut to yield a specimen to include the throat and approximately one half (1/2) inch of the sidewall. One section from each pair of sections (from the same motor) shall be polished, etched and examined under the required magnification. The motor body shall be classed defective if it fails to conform to any of the specified requirements.

4.4.3 Yield strength of motor body. The motor body, after all machining operations have been completed, and prior to anodizing, shall be assembled to the applicable test equipment and subjected to the internal hydrostatic pressure specified for a minimum of 2 seconds. The medium used in testing may be water, or any liquid compatible with the motor body and test equipment. The test equipment shall automatically apply hydrostatic pressures within the prescribed limits. The test equipment shall automatically stamp the component if the component has actually been subjected to the required pressures. The test equipment shall automatically assure that the component is not stamped if it is not subjected to the minimum level specified. The test equipment shall have automatic controls that will lock the equipment in an inoperative condition at any time that the upper level of test pressure is exceeded. Components subject to pressures that exceed the upper specified limit shall be rejected and not returned to any lot. Repair and adjustments to the test equipment shall follow any occurence of excessive pressure. After testing, the motor body shall be dried and visually examined for bulges and deformation. Motor bodies that appear to be deformed beyond drawing dimensions, burst or have threads that fail to hold shall be classed defective



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* 4.4.4 Ultimate strength of motor body. The motor body, after having exhibited acceptable yield strength characteristics, shall be assembled to the applicable destructive test equipment and subjected to an internal hydrostatic pressure of increasing magnitude until the motor body ruptures or until the threads pull out of the test fixture. The pressure level, when failure occurs, shall be recorded to the nearest two hundred (200) pounds per square inch and the results distributed in accordance with the instructions in 6.1.

4.4.5 Contour of motor body throat. The contour of the throat area shall be determined by use of the profile template type measure equipment. The contour shall be determined from a point approximately one half (1/2) inch forward of the radium blending into the cylindrical surface to the rearmost datum diameter in the exit cone. The motor body shall be classed defective if it fails to conform to the limits established by the applicable drawing.

4.4.6 Surface finish of motor body. The surface finish of anodized motor bodies shall be determined by use of "boretracker" type equipment. The surface finish shall be determined by the beginning of the throat approach radius (the radium adjacent to the cylindrical surface) to the rearmost datum diameter is the exit cone. The anodized motor body shall be classed defective if the surface finish fails to conform to the drawing requirement.

4.4.7 Protective coating (salt spray method). The component shall be tested as prescribed in Standard Method B117 in the American Society for Testing and Materials (ASTM) using the required salt concentrations for the required period of time. After testing, the components may be rinsed in warm water, without rubbing, allowed to dry, without wiping, and visually examined. The component shall be classed defective if the visible corrosion exceeds the specified limits.

4.4.8 Protective coating thickness (direct measurement). The motor body shall be sectioned to yield two halves. The primary surfaces of interest are diametrically opposed longitudinal sections containing longitudinal exposure of the contour of the throat. These sections shall include the throat diameter and approximately one-quarter (1/4) inch of the approach surface to the throat. One edge of both specimens (surfaces that were not developed in the same cut on the diameter) shall be polished and etched to reveal the protective coating thickness. This thickness shall be measured to the nearest 0.00005 inch. The fin shall be sectioned perpendicular to its length approximately one-half (1/2) inch from the hole. One of the exposed cross-sections shall be polished and etched to reveal the protective coating thickness. This thickness shall be measured to the nearest 0.00005 inch. The component shall be classed defective if any protective coating thickness fails to conform to the minimum thickness required by the applicable drawing.



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4.4.9 Fin action. The complete assembly, including the fin retainer, shall be placed on an anvil which will locate the rear edge of the exit cone 2.500 inch plus or minus 0.100 inch above the surface of a table and orient the longitudinal axis of the assembly perpendicular to the table. The test equipment should provide for protecting the protective coating from damage during testing. When the assembly is in position for test, the fin retainer is moved towards the threaded end of the motor body thereby releasing the fins. The assembly shall be classed defective if one or more fins fail to fully open (fails to contact the surface which is perpendicular to the longitudinal axis of the assembly and 0.875 ± .015 inches to the rear of the lip of the exit cone).

4.4.10 Physical strength of exit cone. The complete assembly shall be assembled to the applicable test equipment and required stresses applied. The test equipment shall be functionally verified at four hour maximum intervals of production, prior to and immediately following acceptance testing at the beginning and end of the work shift. This verification shall consist of stressing test standards capable of determining that stresses and intended hydrostatic pressures conform to the specified requirements. Stress application measurements shall be centered, longitudinally at the specified locations. The result of each test equipment verification shall be recorded and made available to the Government. The Government may require and witness a verification during any unscheduled, reasonable period of production. Unscheduled verifications shall be kept to a minimum so as to keep interruptions in production to a minimum. If the functional verification reveals that the test equipment is applying non-conforming stress to the product, disposition of the product test since the last acceptable functional verification shall be as follows:

a. Understressed. If one or more stress points are not being adequately stressed, the affected product shall be resubmitted to the test after the required repairs or adjustments have been made to the test equipment.

b. Overstressed. If one or more stress points, including the internal hydrostatic pressure, have been subjected to excessive stress levels, the affected product shall be rejected and removed from the lot and scrapped.

4.4.11 Physical defects. The assembly shall be assembled to the applicable test equipment and the entire inner surface of the exit cone shall be scanned by the probe of an eddy current type tester utilizing the "phase shift" method of detection. Standard motor bodies (standards) containing the specified metallic discontinuities shall be utilized to calibrate and adjust the test equipment to produce the proper accept or reject signal. The test equipment shall automatically mark the assembly only if no reject area or areas were encountered and shall reliably preclude marking of the assembly if one or more areas of non-conformance are encountered.



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4.4.12 Visual defects examination. The motor body of the motor assembly, after the completion of all non-destructive examinations and tests, shall be examined under three (3) power magnification over the complete surface of the entire area between the exit cone lip and approximately 1/4 inch down the cylindrical straight sided portion of the pressure area for compliance with visual defects requirement. The entire remainder of the complete external surface of the motor shall be subjected to unaided visual examination for compliance with the visual defects requirements. The assembly shall be classed defective if it fails to comply with the visual defects requirement on any surface other than the unmachined lug band surface. A motor containing a blemish on the unmachined lug band surface of a nature or visual appearance of depth, or questionability of ultimate depth, beyond .002 inch or beyond the lug band rise thickness, whichever is less, shall be classed defective.

4.4.13 Static functioning. The motor assembly shall be loaded with a production igniter and propellant charge and conditioned at 150 degrees Fahrenheit plus or minus 5 degrees Fahrenheit for a period of 12 hours minimum and statically fired within 2 minutes after removal from the conditioning chamber. A pressure time trace shall be obtained to establish if failure was due to excess pressure (over 10,000 psi). Following firing, the motor shall be visually examined. The motor will be classed defective if visual examination reveals that the motor has fractured or fragmented providing the maximum pressure has not exceeded 10,000 psi.

5. PACKAGING

5.1 Packaging, packing and marking - Level C. The motor parts assemblies shall be packaged, packed and marked in accordance with MIL-STD-1169

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. Data cards shall be prepared for each lot in accordance with the inforamtion specified in MIL-STD-1167.

c. Instructions for distribution of results of ultimate strength tests of motor bodies, and results of first article static firing tests of complete assemblies. Results shall be forwarded to MICOM, DRSMI-RS and ARRCOM, DRSAR-LET, in accordance with appropriate Data Item Description per DODI-5010.12 as directed by Contracting Officer.

d. Provisions for submission of first article sample.



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6.2 Inspection code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Governemnt. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.3).

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the Method of Inspection headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified, and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.



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6.3.3 Submission of Designs for Approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion data for design review will be based on the date of the final submission of designs.

6.4 Submission of first article sample. Instructions as to the location for the evaluation of first article sample shall be obtained for the Contracting Officer. Delivery of test data, hardware and inspection records shall be in accordance with appropriate contract documents.

6.5 Submission of critical defect reports. For U. S. Army procurement, one copy of each Critical Defect report will be furnished the cognizant Government inspection agency.

6.5.1 Critical Defect Reports. The contractor's quality assurance system shall contain a procedure for reporting of Critical defectives found during the course of his inspection and during inspection by the Government. Any Critical Defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those Critical Defects annotated by an asterisk will be furnished. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.

b. Number and revision date of specification.

c. Characteristic code number.

d. Lot or serial number.

e. Number of defectives found.

f. Assignable cause of analysis or occurrence.

g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

6.5.2 Reporting to Other Services. For other than U. S. Army procurement, Critical Defect reporting shall be as prescribed by the Contracting Officer.



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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

Custodian:
Army - MI

Review Activity:
Army - AR

Preparing Activity:
Army - MI

Project No. 1340-A619



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-M-60003D(MI), MOTOR, ROCKET, ASSEMBLY, M54 OR M123 PARTS FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED.

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